Epidemiologic analysis of Crohn disease in Japan: increased dietary intake of n-6 polyunsaturated fatty acids and animal protein relates to the inc...

We examined the correlation between the incidence of Crohn disease and dietary change in a relatively homogeneous Japanese population. The incidence and daily intake of each dietary component were compared annually from 1966 to 1985. The univariate analysis showed that the increased incidence of Crohn disease was strongly ($P < 0.001$) correlated with increased dietary intake of total fat ($r = 0.919$). animal fat ($r = 0.880$), n-6 polyunsaturated fatty acids ($r = 0.883$), animal protein ($r = 0.908$), milk protein ($r = 0.924$), and the ratio of n-6 to n-3 fatty acid intake ($r = 0.792$). It was less correlated with intake of total protein ($r = 0.482$, $P < 0.05$), was not correlated with intake of fish protein ($r = 0.055$, $P > 0.1$), and was inversely correlated with intake of vegetable protein ($r = -0.941$, $P < 0.001$). The multivariate analysis showed that increased intake of animal protein was the strongest independent factor with a weaker second factor, an increased ration of n-6 to n-3 polyunsaturated fatty acids. The present study in association with reported clinical studies suggests that increased dietary intake of animal protein and n-6 polyunsaturated fatty acids with less n-3 polyunsaturated fatty acids may contribute to the development of Crohn disease.